

Supplementary Material: Assessing Decision Reliability in Discrete-Event Simulation

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This document contains supplementary material for the main manuscript “Assessing Decision Reliability in Discrete-Event Simulation: A Quantum-Network Case Study.” It provides the reporting standard and Trust Stack checklist, reviewer and researcher recommendations, the claim reproduction matrix, and the standard event-trace format.

1. STANDARD: A CHECKLIST FOR REPORTING

1.1 Reporting Standard for Bottleneck Diagnoses

We propose that any bottleneck diagnosis should report:

1. **Intervention strength** δ used for each resource.
2. **Number of seeds** and whether common random numbers were used.
3. **Performance metric** (A_{req} or A_{qual}) and fidelity threshold F_{min} .
4. **Queueing model** (concurrent, sequential, or other).
5. **Soft bottleneck vector** p_j with uncertainty (e.g., bootstrap CI), not just the argmax label.
6. **Bottleneck entropy** H_B to indicate whether the label is separable or diffuse.
7. **Degenerate cases** reported separately, not assigned an arbitrary label.

1.2 Trust Stack Checklist

We propose a Trust Stack checklist for reporting simulation-based conclusions. Each row identifies what the manuscript itself fulfills (Table 1).

1.3 Recommendations for Reviewers

Based on the findings of this study, we recommend that reviewers of simulation-based quantum-network papers:

1. **Require semantic transparency.** A paper that uses simulation should document the protocol semantics of the simulator, not just the parameter values. If the semantics are not documented, the paper is not reproducible. The semantic-dimensions table provides a template.
2. **Ask whether the conclusion is structural or model-sensitive.** A structural conclusion (ranking, monotonicity) can be supported by a single simulator. A model-sensitive conclusion (threshold, magnitude) should be validated across simulators. The claim reproduction matrix (Table 2) provides a template for classifying conclusions.
3. **Request cross-simulator comparison for contention-dominated claims.** If the paper claims a capacity-planning result based on a contention-dominated regime, ask whether the result survives a model change. This study shows that 50.8% of configurations show $\geq 4 \times m_c$ disagreement.
4. **Require provenance.** CES scenarios, event traces, and software versions should be archived as supplementary material. A paper without provenance is a claim without evidence.

1.4 Recommendations for Researchers

1. **Report rankings, not just absolute values.** Rankings are more robust across model changes than absolute values. A conclusion that “family A outperforms family B” is more likely to survive a model change than “family A achieves $A_{\text{qual}} = 0.37$.”
2. **Do not compare latency across implementations without harmonizing definitions.** The timing definition must be documented and matched. If the definitions differ, report the difference as a definition

Table 1. Trust Stack checklist. Each row identifies a trust level, what it requires, and what this manuscript fulfills.

Level	Name	Requirement	This manuscript
L1	Physics verification	Analytical benchmarks for physical primitives	B0 gate: harmonized model passes ($\epsilon_{\text{abs}} < 0.02$); WQ fails B0 due to retrieval-in-delivery semantics (documented in the B0 gate)
L2	Semantic transparency	Document all protocol-semantic choices	Semantic-dimensions table: 9 dimensions documented for both systems
L3	Canonical workloads	Run standardized benchmark ladder	B0–B6 ladder (3,950 + 3,840 scenarios), all levels reported
L4	Standard event traces	Export structured event traces	SHA-256 trace hashing for SeQUeNCe/SimQN identity; format in Appendix
L5	Common experiment specification	Accept CES-format input	CES defined and used across all implementations
L6	Model uncertainty reporting	Report model sensitivity alongside point estimates	τ , κ , Δ_m , cluster bootstrap CIs reported throughout

mismatch, not a model disagreement.

3. **Treat capacity-planning conclusions as model-dependent.** Report m_c as a range or with model-sensitivity analysis, not as a single number. This study shows that m_c can differ by $\geq 4\times$ in 50.8% of configurations.
4. **Report the validation level.** State which level of the benchmark ladder (B0–B6) the conclusion claims, and what evidence supports it. A claim at B6 requires cross-model evidence; a claim at B0 does not.
5. **Use cluster bootstrap CIs for cross-simulator comparisons.** Resample configurations (not individual seeds) to avoid pseudoreplication. When CIs include zero, the effect should be reported as not statistically significant.

2. CLAIM REPRODUCTION MATRIX

Table 2 summarizes the three-class taxonomy with the verdict for each claim.

3. STANDARD EVENT TRACE FORMAT

A standard event-trace format enables diagnostic comparison: when two simulators disagree, the event traces can be compared event-by-event to identify where and why the divergence occurs. Table 3 shows the proposed format.

In this study, event traces were used to verify the SeQUeNCe–SimQN bit-identity (SHA-256 hashing across all tested scenarios). The trace format is available in the repository and can be used for event-by-event diagnostic comparison when models diverge.

REFERENCES

Table 2. Claim Reproduction Matrix with three-class taxonomy.

Claim	Verdict	Class	Evidence
C1: Memory-family ranking (4 profiles)	REPLICATED	Structural	$\tau = 1.0$, perfect ranking agreement
C2: Topology ranking	REPLICATED	Structural	$\tau = 1.0$, perfect ranking agreement
C3a: Monotonicity of $A(m)$	REPLICATED	Structural	100% monotonicity both models
C3b: Diminishing returns	UNRESOLVED	—	B6 m grid too coarse
C4a: m_c regime-dependence	STRUCTURALLY REPLICATED	Structural	Both agree which families need more memory
C4b: m_c numerical values	MODEL-SENSITIVE	Model-sensitive	50.8% $\geq 4\times$ disagreement
C5: Scalar collapse failure	STRUCTURALLY REPLICATED	Structural	$R^2 < 0.2$ both models
Latency ranking	NOT COMPARABLE	Definition-sensitive	$\tau = -0.29$, timing definition mismatch
Finite-memory contention	NOT REPLICATED	Model-sensitive	$\tau = 0.17$ in B4

Table 3. Proposed standard event-trace format.

Field	Description
timestamp	Simulation time of event
scenario_id	CES scenario identifier
request_id	Unique request identifier
node_id	Node where event occurs
event_type	generate, swap, deliver, reject, timeout, ...
memory_slot	Memory slot index (if applicable)
link_id	Link identifier (if applicable)
fidelity_before	Fidelity before event
fidelity_after	Fidelity after event
resource_state	Memory occupancy, queue length
cause	Reason for event (if applicable)